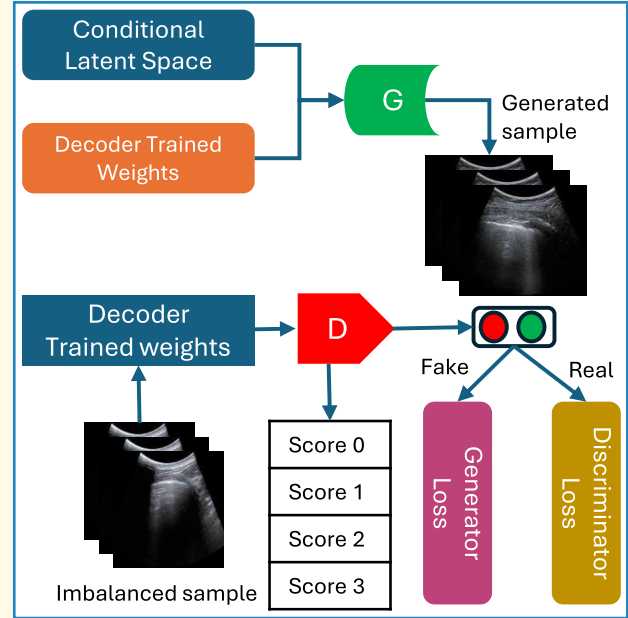


Synthetic Lung Ultrasound Data Generation Using Autoencoder With Generative Adversarial Network

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Abstract—Class imbalance is a significant challenge in medical image analysis, particularly in lung ultrasound (LUS), where severe patterns are often underrepresented. Traditional oversampling techniques, which simply duplicate original data, have limited effectiveness in addressing this issue. To overcome these limitations, this study introduces a novel supervised autoencoder generative adversarial network (SA-GAN) for data augmentation, leveraging advanced generative artificial intelligence (AI) to create high-quality synthetic samples for minority classes. In addition, the traditional data augmentation technique is used for comparison. The SA-GAN incorporates an autoencoder to develop a conditional latent space, effectively addressing weight clipping issues and ensuring higher quality synthetic data. The generated samples are evaluated using similarity metrics and expert analysis to validate their utility. Furthermore, state-of-the-art neural networks are used for multiclass classification, and their performance is compared when trained with GAN-based augmentation versus traditional data augmentation techniques. These contributions enhance the robustness and reliability of AI models in mitigating class imbalance in LUS analysis.

Index Terms—Data augmentation, deep convolutional (DC) discriminator, generative adversarial network (GAN), lung ultrasound (LUS), multiclass classification, supervised autoencoder (SA).



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I. INTRODUCTION

THE utilization of ultrasound technology is steadily growing as it aids in both the diagnosis and surveillance of various lung ailments such as fibrosis and pneumonia. Within the spectrum of pneumonia, numerous types exist; however, our study concentrates specifically on viral pneumonia. Notably, viral pneumonia is attributable to influenza viruses, respiratory syncytial virus (RSV), and SARS-CoV-2 (the virus responsible for COVID-19) [1].

Lung ultrasound (LUS) stands as a pioneering technology in medical diagnostics due to its cost-effectiveness, absence of radiation, and reliable nature. Its pivotal role in clinical assessments, delineating lung surface irregularities such as pleural line irregularities, artifacts, and consolidation areas, underscores its significance [2], [3]. To assess these LUS patterns, clinicians score LUS data by visually interpreting ultrasound images based on a four-level scoring system from score 0 to score 3 [4]. Score 0 represents a healthy lung surface showing a continuous pleural line with horizontal artifacts. Score 1 indicates the presence of vertical artifacts alongside an indented pleural line. Score 2 signifies small consolidation areas appearing beneath a fragmented pleural line. Finally, score 3 indicates the presence of a white lung,

Highlights

- **This study proposes a novel method for automated generation data based on the generative model. Specifically, we introduce a supervised autoencoder generative adversarial network (SA-GAN) with gradient penalty loss function.**
- **With the help of the autoencoder, we resolve the mode collapse issues after building the conditional latent space by utilizing the train weights of the autoencoder.**
- **Results showed that our proposed technique outperformed the state-of-the-art results in terms of the synthesis data generation which further evaluated through expert assessment analysis and deep learning classifiers.**

with or without consolidations, covering at least 50% of the pleural line. Unfortunately, this procedure is highly subjective and leads to a poorly reproducible methodology. To tackle this shortcoming, artificial intelligence (AI) systems can be used for standardizing LUS data analysis [5].

To train deep learning (DL) models, the dataset needs to be generally diverse (e.g., acquired with different ultrasound scanners and by different operators) and balanced in terms of class distribution. However, while diversity can be easily achieved by means of multicentre clinical studies, generating a large and balanced dataset is particularly challenging. This is particularly true in medical imaging since the patterns connected to the most severe cases (white lung with consolidation areas) are generally found in a minority of the analyzed patients. This poses the problem of how to increase the performance of DL models while disposing of imbalanced datasets [6].

Traditional augmentation techniques, including oversampling and synthetic sampling, have shown limitations in handling complex datasets such as LUS analysis [7]. Methods such as SMOTE can produce duplicates or noisy samples, affecting the overall dataset distribution. Recent advancements propose techniques such as geometric SMOTE (G-SMOTE) that operate within the input space, yet these approaches still fall short in addressing the complexity of medical image analysis [8].

In the field of medical imaging, many researchers use the advanced DL approaches such as generative adversarial network (GAN) for data augmentation [9]. GAN is considered the best generative AI model to generate synthetic images, especially in minority classes [10]. The basic architecture of the GAN is based on two DL models, generator (G) and discriminator (D) [11]. Various GAN approaches have been used for medical applications. Frid-Adar et al. [12] used the deep convolutional (DC) GAN to enrich the limited dataset of computer tomography (CT) liver lesions. However, they trained separate GANs for each lesion class which increased the training complexity and the quality of the generated sample was suboptimal. In addition, their proposed approach does not generate multiclass samples together which would be more worthwhile. For anomaly detection, Schegl et al. [13] proposed the AnoGAN, with a unique approach for mapping the image space to latent space, for new data generation based on images acquired from healthy subjects. However, the approach is limited to the quantitative analysis of the subgroup of anomalies.

To translate the patches of magnetic resonance imaging (MRI) brain images to CT images, Nie et al. [14] used the 3-D GAN by concatenating several GANs. However, it is worth noting that this approach carries computational complexity, and interpretability regulatory validation challenges, which need to be carefully addressed for its successful adoption in clinical practice. To handle the imbalanced dataset of chest X-ray images and to increase the training samples, Salehinejad et al. [15] used a DC-GAN for the generation of chest X-rays that mimic common chest pathologies data augmentation for liver lesion classification. Their approach generated the images in minority classes, but the quality of the images was limited.

On the other hand, literature on GAN technology specifically for ultrasound images is very limited. Pang et al. [16] used the GAN to generate high-quality synthetic breast ultrasound mass images, improving radiomics feature extraction from unlabeled data, and enhancing breast cancer classification performance. However, the generated samples exhibit poor noise. These limitations in image quality could impact the model's overall effectiveness in terms of generalizability. Douzas et al. [8] and Liang et al. [17] also used the pixel-based GAN approaches for the generation of high-resolution images from low-resolution images on three different datasets. The results positively impacted the performance of DL models.

After a comprehensive review of the current state-of-the-art in the field, it has come to our attention that numerous researchers have used generative adversarial network (GAN) technology on CT images, X-ray images, and MRI, whereas only a few studies were on ultrasound images. In particular, only one study [17] specifically addressed the application of GAN for LUS without addressing the problem of class balancing. For this reason in this study, we introduced a novel method based on a supervised autoencoder generative adversarial network (SA-GAN) for data augmentation of LUS to handle the class balancing issues in minority classes which were missing in the state-of-the-art.

Specifically, our objectives can be summarized as follows.

- 1) Oversampling the minority classes with unique LUS patterns to resolve the class unbalancing issues through SA GAN.
- 2) Investigate the generated data characteristics through similarity measurements between GAN-generated and real LUS images.
- 3) Compare the performance of multiclass classification models when trained with GAN-based or classic data augmentation.

TABLE I

INFORMATION ON THE HOSPITALS, PATIENTS, AND FRAME DISTRIBUTION IN THE TRAIN AND TEST SPLITS OF THE LUS PNEUMONIA DATASETS USED IN THIS WORK (THE BOLDFACE VALUES REPRESENT THE SUM OF THE VALUES FOR EACH COLUMN IN THE TRAIN AND TEST)

	Hospital	No# of Patients	No# of frames (Score 0)	No# of frames (Score 1)	No# of frames (Score 2)	No# of frames (Score 3)	Total No# of frames
Train	Brescia	10 COVID-19	607	1302	1147	1311	4367
	Rome	3 (1 COVID-19, 2 healthy)	663	-	25	21	709
	Lucca	4 (2 COVID-19, 2 healthy)	330	72	22	-	424
	Total	17	1600	1374	1194	1332	5500
Test	Brescia	2 COVID-19	405	406	295	280	1386
	Rome	1 COVID-19	-	2	19	23	44
	Pavia	1 COVID-19	30	16	3	37	86
	Total	4	435	424	317	340	1516

II. MATERIALS AND METHODS

A. Dataset Selection

1) *Dataset-v1*: Our method involved a subset of convex image frames from Italian COVID-19 Lung Ultrasound Database (ICLUS-DB) [18]. This dataset comprises 58 924 image frames extracted from 277 videos, collected from 35 patients diagnosed with COVID-19 pneumonia. These image frames were categorized into four classes based on the standard scoring system outlined by Soldati et al. [4]. For training, we initially used a smaller set of convex images from 17 patients, including 13 COVID-19 and four non-COVID-19 cases. The test set comprised images from four patients with COVID-19. A total of 5500 LUS image frames were used for GAN training, resulting in a dataset with class imbalance: 1600 score 0 frame, 1374 score 1 frame, 1194 score 2 frames, and 1332 score 3 frames. For testing, 1516 image frames from four patients were used, distributed as follows: 435 score 0, 424 score 1, 317 score 2, and 340 score 3. A detailed breakdown of the dataset distribution is provided in Table I.

2) *Dataset-v2*: Dataset-v2 represents the entire ICLUS-DB, presented in [18]. It includes 277 LUS videos from 35 patients, corresponding to 58 924 frames (45 560 acquired with convex probe and 13 364 frames with linear probe). Among the 35 patients, 17 of them (49%) were confirmed COVID-19 positive using RT-PCR test, four patients (11%) were COVID-19 suspected, and 14 (40%) were healthy and symptomless individuals. The data were acquired from Fondazione Policlinico Universitario San Matteo IRCCS, Pavia, Italy, Fondazione Policlinico Universitario A. Gemelli IRCCS, Rome, Italy, Valle del Serchio General Hospital, Lucca, Italy, Brescia Med, Brescia, Italy, and Tione General Hospital, Tione, Italy. Based on the scoring system proposed, a total of 58 924 video frames were divided into four different categories. The training set consists of 48 215 image frames, while the test set contains 10 709. The training set is imbalanced, with 48 215 samples for score 0, 11 147 for score 1, 15 883 for score 2, and 4523 for score 3. The test set includes 3311 samples for score 0, 3148 for score 1, 3089 for score 2, and 1161 for score 3.

B. GAN-Based Data Augmentation

This study introduced a novel method based on an SA-GAN. The SA-GAN latent space is built with the help of an autoencoder. Therefore, we first discussed the architecture of the autoencoder [19], [20]. Autoencoder works with unconditional manners and can easily handle mode collapse issues. The basic architecture is based on an encoder (E_n), latent vector (L_v), and decoder (D_e) (see Fig. 1). The E_n translates the images

into L_v and D_e translates the given L_v to reconstruct the images.

1) *Generator/Decoder*: As shown in Fig. 1, D_e takes the latent dimension (I_{dim}) as input and reconstructs the image based on three convolutional 2-D transform layers. Each layer used the following hyperparameters: filter value, kernel size (4×4), strides (2×2), padding, and kernel initializer. After each layer, batch normalization and the LeakyReLU activation function [21] are used. This activation function helps D_e effectively learn and model the complex mapping from the latent space to the output image. The formula of leakyReLU is given below

$$f(k) = \begin{cases} k, & \text{if } k \geq 0 \\ \alpha \cdot k, & \text{if } k < 0 \end{cases} \quad (1)$$

where k represents the input to the LeakyReLU function, and α is a small constant that determines the slope of the function for negative values, preventing the complete inactivation of neurons during training. This LeakyReLU activation function is applied elementwise to each value in the output tensor after the convolutional layers in the decoder network. The output from the last layer with I_{dim} is further used to build D_e architecture.

The extra layer based on the embedded model is created to obtain Cl_v . The initial layers of the embedded model consist of C_l , the number of classes (n), and I_{dim} as input parameters, which are further fed to flatten layer (x). In last I_{dim} with C_l as a new layer ($y \rightarrow (I_{\text{dim}} \times C_l)$) is concatenated. On the other hand, the flatten layer (x) and new layer (y) are concatenated to build the embedded model. The train weights of D_e and the embedded model are further used for training the generator model.

2) *Discriminator/Encoder*: The D followed the conditional Wasserstein GAN (CW-GAN) architecture [22], which is similar to E_n with the only difference consisting of receiving as inputs both the images and corresponding labels. D architecture is the same as a DC neural network (DCNN) that uses four DC 2-D layers together with learning parameters such as kernel size (4×4), strides (2×2), padding, and kernel initializer (see Fig. 1). Each layer used the LeakyReLU activation function. The output of each layer is directly fed to the flatten layer (x). The other flatten layer (y) used the number of class and label information which concatenated together ($x \times y$) as a new dense layer $d(x \times y)$. The features are taken from the trained weights of E_n excluding the output layer as feature maps and concatenated with C_l as a new dense layer that gives the validity to classify each image into real or fake and with respect to different classes.

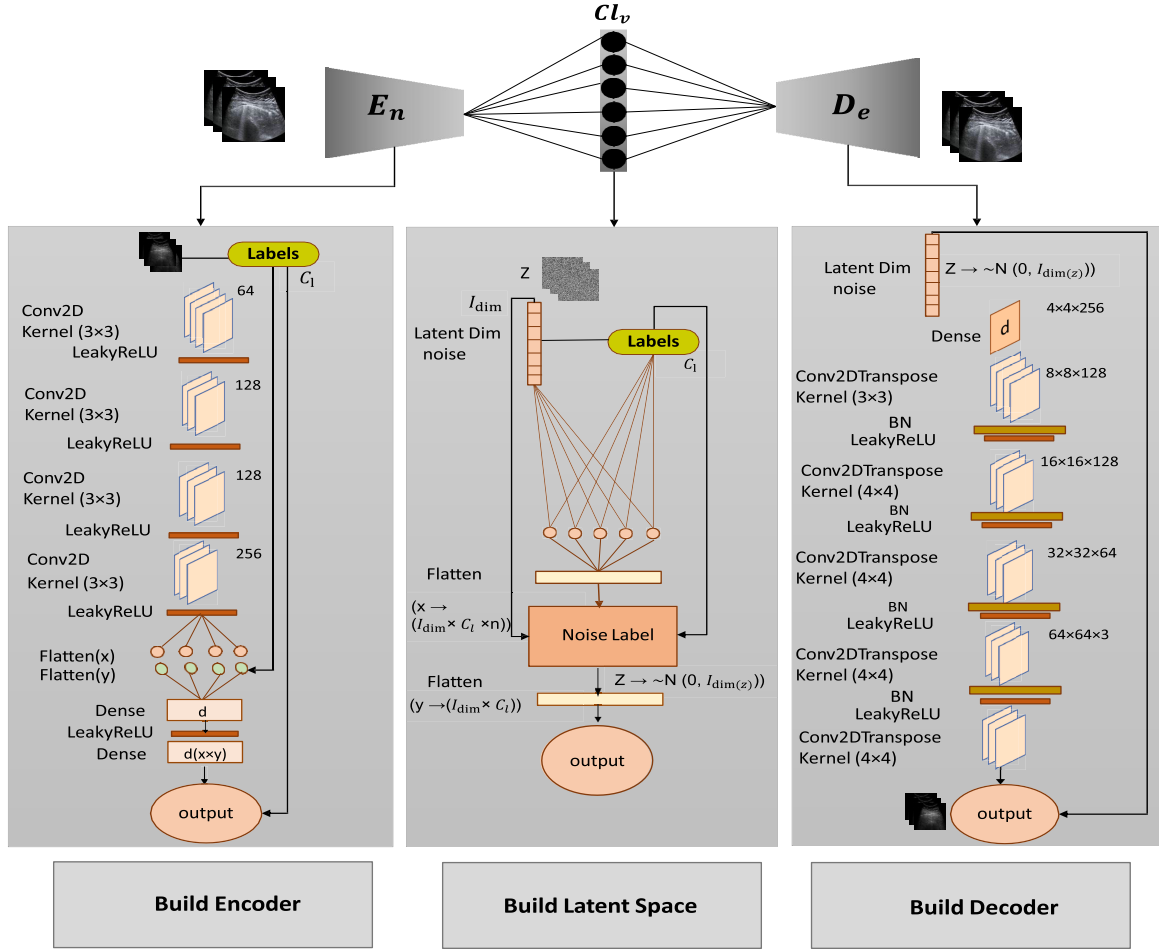


Fig. 1. Architecture of SA. The first block (left) represents the batch of convolutional layers for building the encoder model in conditional manners, whereas the second block (center) shows latent space which is created through latent dimension (l_{dim}) with Cls' information to build the conditional latent space (Cl_v). The last block (right) shows the architecture of the decoder. In this case, The network used l_{dim} with a batch of convolutional 2-D transpose layer, which further helps G for image reconstruction.

3) Loss Function: During the adversarial training of SA-GAN (see Fig. 2), G 's responsibility is to generate the synthetic images in minority classes. On the other hand, D followed the DCNN architecture. The training feedback of SA-GAN is based on GP_{loss} [23]. This penalty encourages the Lipschitz continuity of the D function, which means it makes D 's gradients well-behaved. Ensuring Lipschitz continuity prevented issues such as mode collapse and vanishing gradients. It ensures that D provides gradients even when it is confident about its decisions, helping G continue to learn. Also by penalizing sharp gradients in D , the customized loss function helps D to provide meaningful gradients to G even when G 's output is far from data distribution. This mitigates the risk of mode collapse, where G focuses on a few modes of data distribution, as shown in the following equations:

$$D_{loss}(X_r, Z, L_r, L_f, L_w) = -E_{(x_r, l_r) \sim (X_r, L_r)} [\log(D(x_r, l_r))] - E_{(z, l_f) \sim (Z, L_f)} [\log(1 - D(G(z, l_f), l_f))] - E_{(x_r, l_w) \sim (X_r, L_w)} [\log(1 - D(x_r, l_w))] + \lambda E_{(\tilde{x}, l_r) \sim (X_r, L_r)} [\log(1 - D(x_r, l_w))] \quad (2)$$

$$G_{loss}(Z, L_f) = -E[\log(D(G(z, l_f)))]. \quad (3)$$

In (2), D_{loss} represents the discriminator loss function which is calculated through the real images (X_r), where x_r is a sample of the real distribution X_r . Z is the random noise vector, L_r represents the real labels of each single label l_r , fake images are represented as X_f , which are produced by G , and each sample of the fake label (L_f) distribution is represented as l_f . To enhance the learning of D , the extra cross-entropy loss function of the wrongly classified label (L_w) is also used, where l_w represents a sample from L_w , which is randomly selected from L_r , and it is directly fed to D . E represents the expected value. In (2), λ is the hyperparameter, and to handle the weight clipping issues, x is used to calculate the gradient norm between X_r and X_f . The gradient norm of D output is relative to its input is calculated after feeding $\tilde{x}$ into the critic D . In our case, D provided the four classification types, which categorize each single image into real or fake and specific classes (score 1, score 2, and score 3). To analyze the performance of D model, the initial loss (I_{loss}) is calculated between l_r , l_f , and l_w , whereas GP_{loss} is calculated between x_r , x_f , and l_w . The I_{loss} and GP_{loss} are combined together to obtain the final loss, as shown in the following equation:

$$D_{loss} = I_{loss} + GP_{loss}. \quad (4)$$

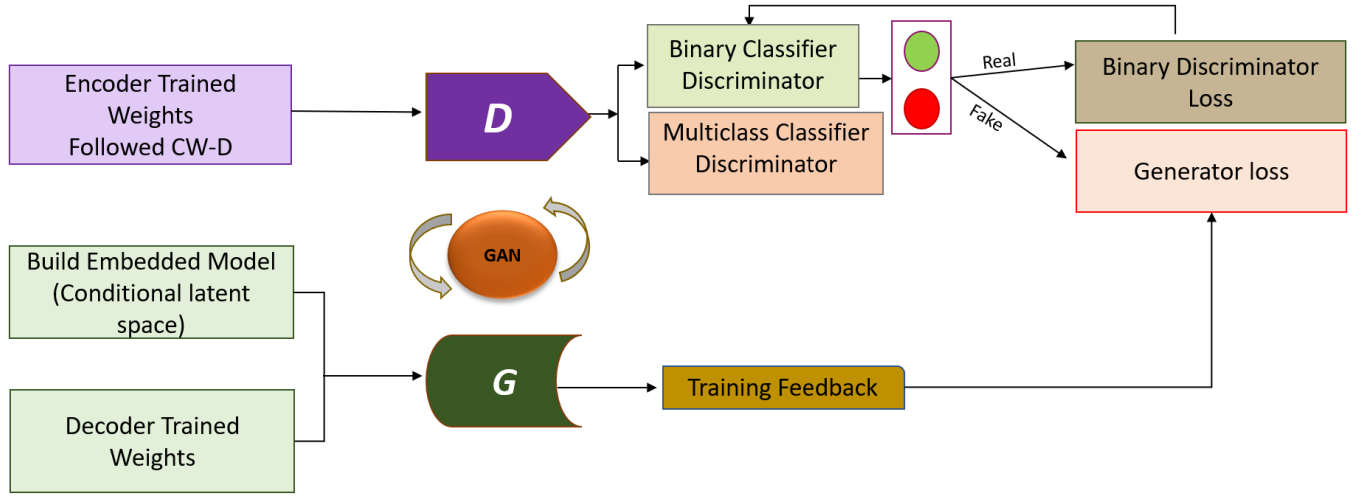


Fig. 2. Architecture of SA. The first block (left) represents the batch of convolutional layers for building the encoder model in conditional manners, whereas the second block (center) shows latent space where we have taken the latent dimension (l_{dim}) with Cls' information to build the conditional latent space (Cl_v). The last block (right) shows the architecture of the decoder. In this case, we used l_{dim} with a batch of convolutional 2-D transpose layer, which further helps G for image reconstruction.

TABLE II
HYPERPARAMETERS OF SA-GAN

Parameters	Values
Batch Size (autoencoder)	4
Epochs (autoencoder)	500
patience (autoencoder)	10
loss function (autoencoder)	mean absolute error
latent dimension	128
SEED	42
Optimizer	Adam
Learning rate	0.0002
β_{a1}	0.5
β_{a2}	0.9
Activation Function (autoencoder)	LeakyReLU
Batch Size (GAN training)	128
patience (GAN training)	20
Epochs (GAN training)	150
Learning Steps/ Iterations (GAN training)	1500
D_{loss}	Sigmoid Cross-Entropy Loss (for L_r , L_f , and L_w), GP_{loss}
G_{loss}	Sigmoid Cross-Entropy Loss (to classify fake images as real).

G_{loss} is calculated between X_f and L_f as shown in the following equation:

$$G_{loss} = (X_f, L_f) \text{ where } X_f \leftarrow (Z, L_f). \quad (5)$$

G takes random noise vector Z as input, where $Z \leftarrow \sim N(0, I_{dim(z)})$ to generate the images which not only is realistic but also preserves the underlying structure and distribution of the real data. G is trained on the minority classes to generate a balanced set of data. It automatically saved the generated images into different classes with the help of class label (Cl) information. In (4) and (5), D_{loss} and G_{loss} are the final losses that are further used for SA-GAN training which helps in stabilizing the training process without weight clipping. The structure of SA-GAN is described in Fig. 2. The summary of the model's architecture is given in https://drive.google.com/file/d/lxTi297Sowm_yCc3a_N_dslzO6Fys_3AC/view?usp=drive_link.

SA-GAN is trained iteratively through a competitive loop between the generator and the discriminator. The generator creates synthetic images conditioned on Cls and noise input from the latent space, while the discriminator evaluates these

images to differentiate between real and synthetic data. Based on the feedback provided by the discriminator, the generator refines its outputs in subsequent iterations. Over time, this iterative process enables the generator to progressively learn and replicate the underlying data distribution for each class, resulting in increasingly realistic and diverse synthetic samples. The hyperparameters used for model training are mentioned in Table II.

C. Classic Transformation-Based Data Augmentation

The second technique is based on the classic method named affine transformation, which is also used by benchmark study [18]. The duplicate copies of original images are created by applying different transformations include translation (max = $\pm 15\%$), rotation (max = $\pm 15^\circ$), scaling (max = $\pm 45\%$), shearing (max = $\pm 4.5^\circ$), horizontal flipping ($p = 0.5$), multiplication with a constant (max = $\pm 45\%$), contrast distortion (max = $\pm 45\%$), Gaussian blurring ($\sigma_{max} = 3/4$), and additive white Gaussian noise ($\sigma_{max} = 0.015$). The augmentation is applied on minority classes to balance with the majority class that contains the highest number of images.

D. Evaluation Metrics

The performance of the models is evaluated through confusion matrices. With the help of the Keras, Sklearn libraries [24], classification report is based on the following metrics: TP, TN, FP, and FN indicate true positive, true negative, false positive, and false negative, respectively. The above-mentioned metrics have been computed for each class c (scores 0, 1, 2, and 3)

$$\text{Accuracy}_c = \frac{\text{TN}_c + \text{TP}_c}{\text{TN}_c + \text{FP}_c + \text{TP}_c + \text{FN}_c} \quad (6)$$

$$\text{Precision} = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c} \quad (7)$$

$$\text{Recall} = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \quad (8)$$

$$\text{F1 Score}_c = 2 \times \frac{\text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (9)$$

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}. \quad (10)$$

F1-score (also known as F-measure, or balanced F-score) is a metric which measures model performance by calculating the harmonic mean of precision and recall for the minority positive class. The structural similarity index (SSIM) measures the similarity between two images, where μ_x and μ_y are the mean intensities, σ_x^2 and σ_y^2 are the variances, σ_{xy} is the covariance, and C_1 and C_2 are constants for numerical stability.

The whole experiment was performed on a PC with an anaconda Spyder tool using TensorFlow Python API supports Python 3.7 and Keras as DL APIs. We used NVIDIA GeForce RTX 3060 GPU, CUDA 10.1, 16-core Processor, and RAM 64 GB.

III. RESULTS

We began by conducting experiments on a smaller dataset (see Section II-A1) to generate images for minority classes using SA-GAN. The primary objective was to assess the quality of the generated images and determine their suitability for further analysis. Thus, Dataset-v1 was used solely for evaluating the SA-GAN approach. Initially, SA-GAN was used to generate images, followed by statistical analyses to evaluate how closely these generated images resembled real ones. At this stage, the focus was on similarity analysis and a blind review by three LUS experts to assess their ability to differentiate between real and generated images.

In contrast, Dataset-v2, which is larger and multicenter, was used for a more comprehensive analysis and to facilitate a fair comparison between SA-GAN and classic transformation methods for class balancing.

A. Results of SA-GAN-Based Data Augmentation on Dataset-v1

SA-GAN is trained on several iterations. After each iteration, we saved the 900 (score 1: 226, score 2: 406, score 3: 268) LUS samples generated in minority classes. The quality of images improves gradually after each iteration. For this reason, we have trained our model for different iterations.

We initially saved and analyzed the generated images after five iterations (learning step). The results show that our model needed more training. We thus trained our model on 100 iterations and saved the generated images with respect to their corresponding classes. The results demonstrate that LUS image artifacts are now generated. For this reason, we continued the training process and saved the reconstructed images after 500 iterations. A significant improvement in image quality can be observed with 500 iterations. This shows that image quality can be improved by increasing the number of iterations. We then continued our training process until 1500 iterations where the maximum resolution is achieved. As examples, synthetic LUS images generated after five, 100, 500, and 1500 learning steps (iterations) are displayed in Fig. 3.

1) *Similarity Measurements*: The study compared 900 GAN-generated images to a set of 5500 real images used for training. It assessed their similarity in three different score categories. For each score, similarity between generated and real images was measured, showing over 50% similarity across all scores: 58% for score 1, 57% for score 2, and 58% for score 3.

In addition, 5500 unseen real images were compared with the 900 generated images, revealing median similarity values of 56% for score 1, 56% for score 2, and 57% for score 3.

Finally, comparisons were made between training data and unseen data, displaying median similarity values of 65% for score 1, 63% for score 2, and 64% for score 3. All these findings are graphically represented in Fig. 4.

Alternatively, similarity analysis was performed within each class, examining real images against one another, generated images against their counterparts, and unseen images against other unseen ones (refer to plots D–F in Fig. 4). For real versus real images, the similarity levels were 69%, 70%, and 72% for scores 1–3, respectively. Comparisons between generated images showed similarities of 74%, 74%, and 70% for scores 1–3. Finally, unseen images compared with other unseen ones exhibited similarities of 71%, 74%, and 70% for scores 1–3.

2) *Expert Assessment*: This analysis used a binary classification approach, where 0 indicated that clinicians could not identify whether the images were real or generated, and 1 represented correct identification. A total of 100 images (50 GAN-generated and 50 real) were randomly selected from four classes, and three expert clinicians evaluated these images to classify them as real or fake. The clinicians' classifications were compared against the ground-truth labels. To ensure objectivity, the clinicians were blinded and had no prior exposure to the real or generated images. The results show that Clinician 1 (Tiziano Perrone), Clinician 2 (Andrea Smargiassi), and Clinician 3 (Riccardo Inchingolo) achieved classification accuracies of 60%, 42%, and 53%, respectively. However, the overall Fleiss' kappa agreement among clinicians suggests a significant level of disagreement; see Table III.

B. Results of SA-GAN and Classic Data Augmentation on Dataset-v2

After analyzing the SA-GAN performance on the smaller dataset, we extended our analysis to the entire dataset represented as Dataset-v2 (see Section II-A2). In this phase,

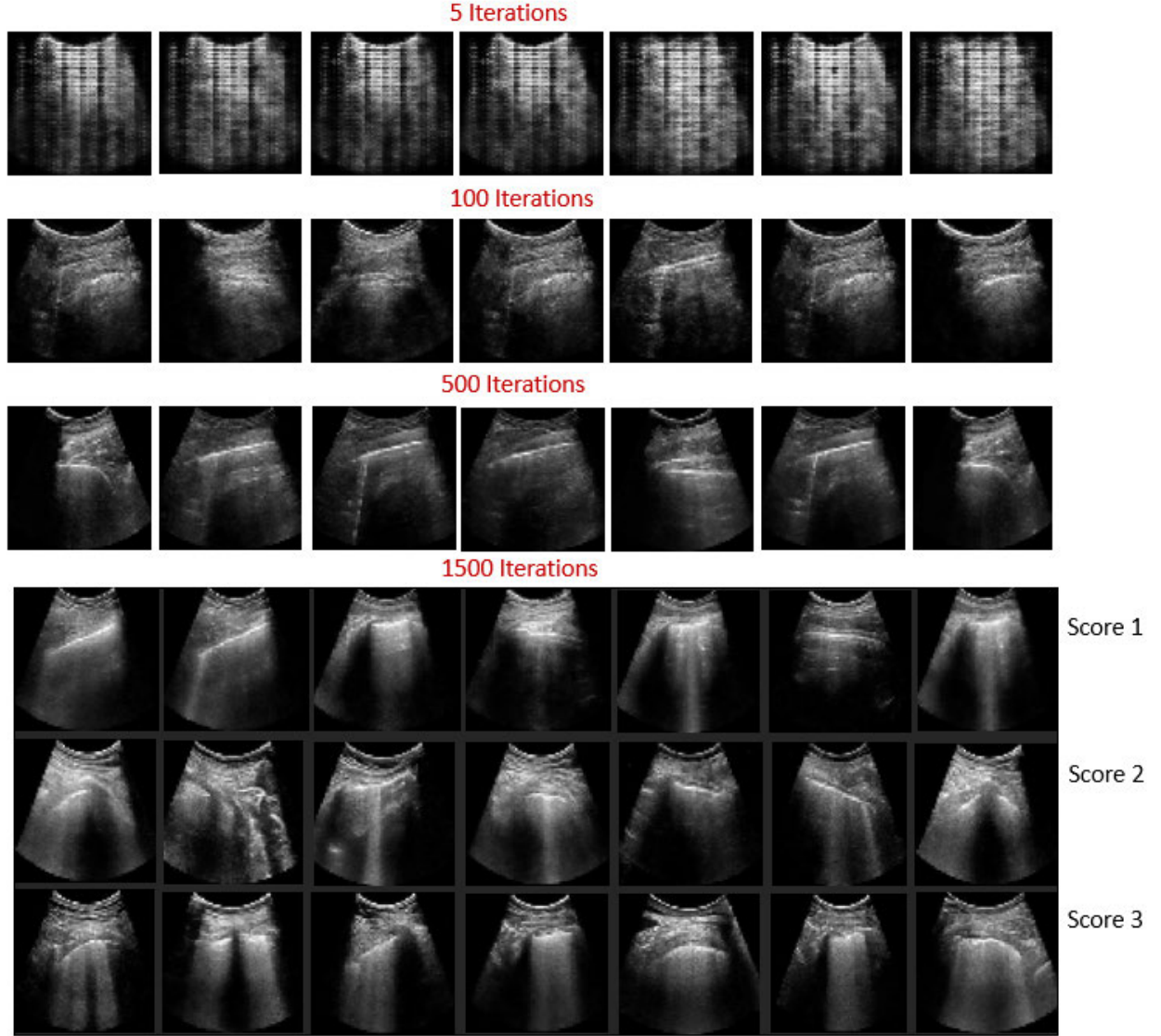


Fig. 3. Samples generated by the SA-GAN for the minority classes, identified as score 1, score 2, and score 3, visually demonstrate that the model can produce distinct features. This capability effectively balances the dataset by leveraging the smaller training set represented as Dataset-v1.

TABLE III
INTRRATER AGREEMENT

Kappa	Raters	Agreement
Cohens Kappa	TP and AS	-0.113 (95% CI = -0.232 - 0.005)
Cohens Kappa	TP and RI	0.167 (95% CI = -0.036 - 0.369)
Cohens Kappa	AS and RI	0.134 (95% CI = -0.041 - 0.309)
Fleiss' Kappa	TP and RI and AS	-0.007 (95% CI = -0.120 - 0.106)

we applied both the SA-GAN and classic transformation to oversample the minority classes. The balanced sets were then used to assess the performance of DL classifiers and to determine which technique is more effective for class balancing. The subsequent sections provide detailed results of these evaluations.

1) *SA-GAN-Based Data Augmentation*: The SA-GAN model was trained using the training set of Dataset-v2, with model checkpoints saved after each iteration. Early stopping criteria were implemented to generate synthetic samples based on the generator's minimum loss (D_{loss} : 1.1079, G_{loss} : 1.8606),

with the checkpoint patience set to 20. These checkpoints enabled the selection of optimal model weights, ensuring minimal loss and enhanced stability. Using these trained weights, we generated a total of 18433 samples for the minority classes: 5515 images for score 1, 779 images for score 2, and 12139 images for score 3 (see Fig. 5). This balanced distribution, incorporating synthetic samples generated by SA-GAN and real samples (trainset of Dataset-v2), is referred to as Trainset-1.

2) *Classic Transformation-Based Data Augmentation*: We also applied a series of classic transformations (detail description in Section II-C) to augment the minority classes in the training set of Dataset-v2. After applying these transformations, we generated a total of 18433 images: 5515 for score 1, 779 for score 2, and 12139 for score 3 (see Fig. 6). This balanced distribution, incorporating samples obtained from classic transformations and real samples (trainset of Dataset-v2), is referred to as Trainset-2.

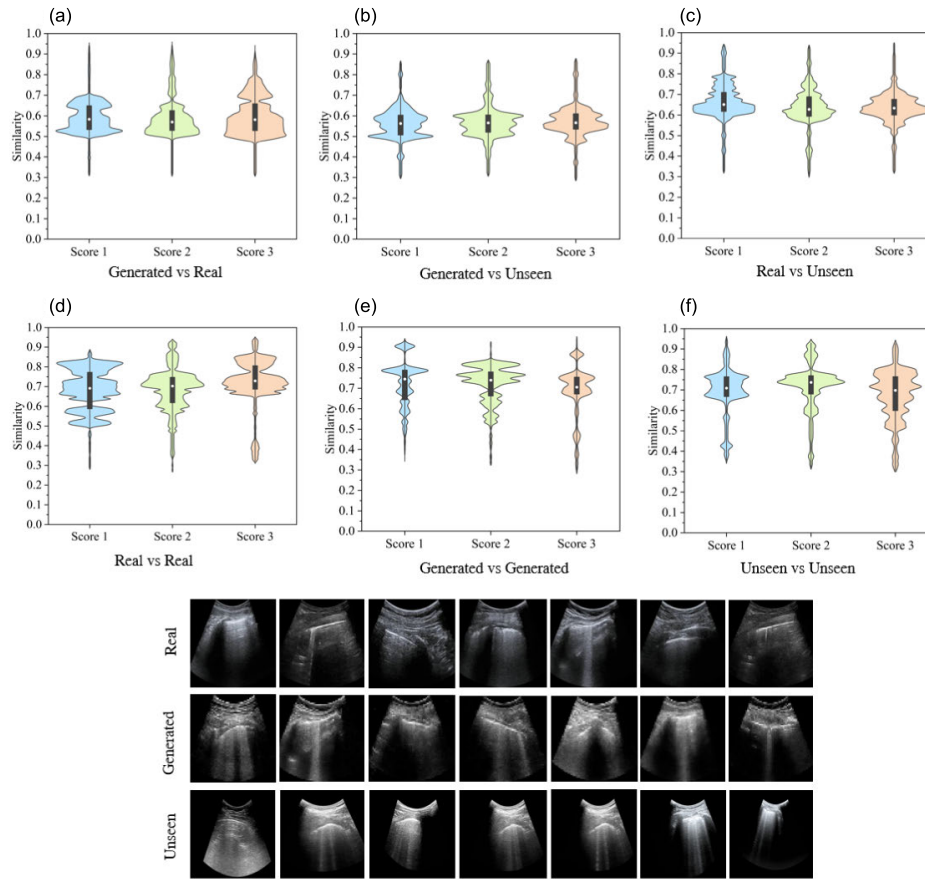


Fig. 4. Violin graphs of similarity measurements. The x-axis of each graph represents the three minority classes (scores 1–3). The y-axis represents the similarity value. Graph A represents the scorewise similarity between each single generated image with all real images (images that used for SA-GAN training). Graph B shows the similarity between the SA-GAN generated images with real unseen images. Graph C illustrates the similarity between real and unseen LUS images. Graphs D–F illustrate the comparison of similarity within identical classes for real versus real, generated versus generated, and unseen versus unseen images. The graphical images at the bottom also depict the real, generated, and unseen (not used for model training) images that used for similarity measurements.

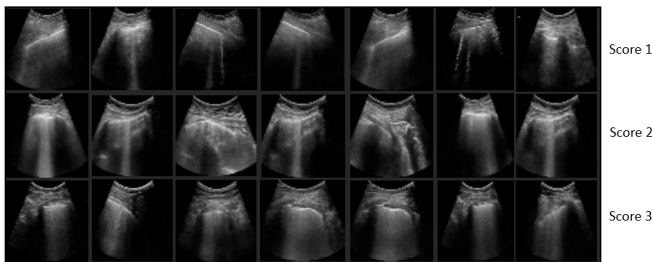


Fig. 5. Samples generated by SA-GAN using the multicenter dataset, referred to as Dataset-v2. The examples of generated artifacts for the minority classes scores 1–3 demonstrate the model's ability to produce synthetic samples by training on a larger dataset.

3) Performance Evaluation of DL Classifiers: Both the training sets (Trainset-1 and Trainset-2) are then used for DL models primarily to evaluate the impact of data augmentation. For this reason, we applied the series of neural networks for multiclass classification that begins with a DC discriminator model (DCDM). This model incorporates the same layers as the proposed discriminator model, except the final dense layer, which consists of four units designed for four levels of scoring [4]. We also implemented a DCNN with multiple

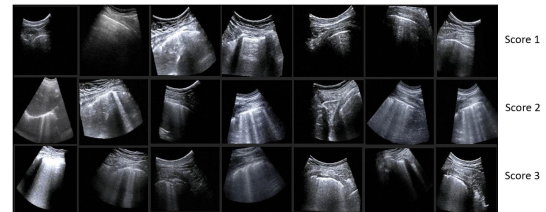


Fig. 6. Samples generated through classic transformations using the multicenter dataset, designated as Dataset-v2. The examples of augmented samples for the minority classes scores 1–3 are presented rowwise.

convolutional layers, BatchNormalization, and activation functions. In addition, we leveraged pretrained ImageNet models, including VGG-16 and VGG-19, alongside the state-of-the-art ResNet-18 model [25] outperforming for LUS pattern classification. Unlike the pretrained models, ResNet-18 was trained from scratch without using pretrained weights. All these networks were trained with categorical cross-entropy loss and the Adam optimizer, with validation carried out on the test set of Dataset-v2.

Fig. 7 illustrates the performance of the DL models using Trainset-1. The matrix reveals that all the classifiers exhibit

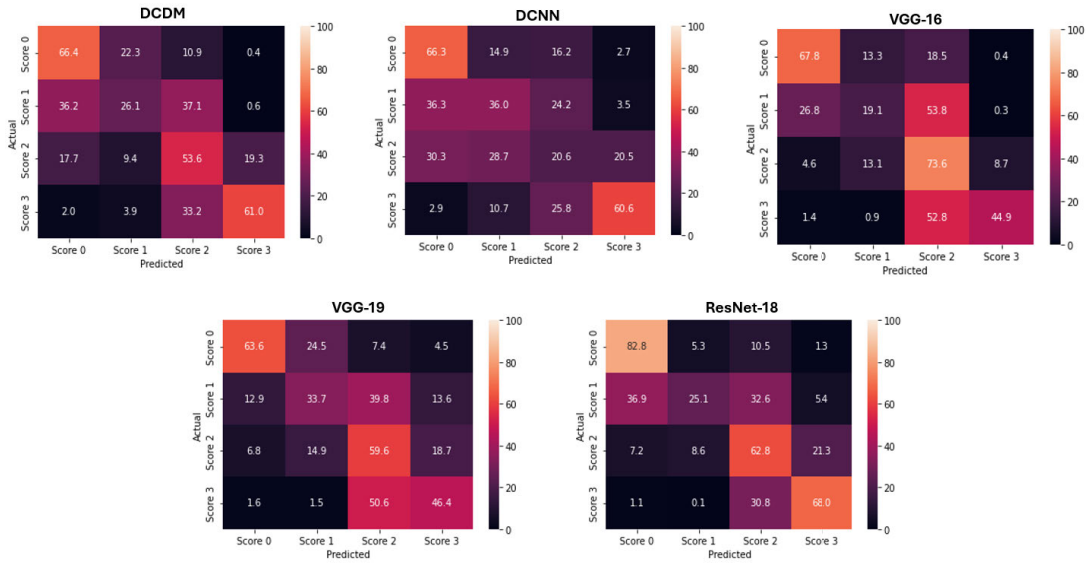


Fig. 7. Confusion matrix displays the performance of DL classifiers based on four metrics, using Trainset-1 (SA-GAN and real data).

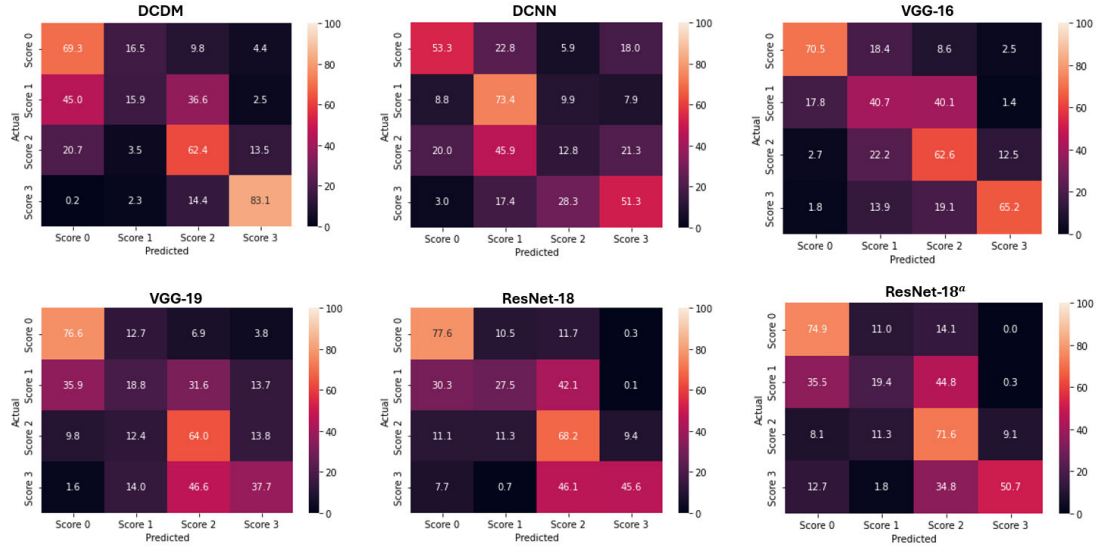


Fig. 8. Confusion matrix displays the performance of DL classifiers based on four metrics, using Trainset-2 (affine transformation and real data).

relatively low performance for score 1 (vertical artifacts), while true positive rates are notably higher (over 60%) for score 0 (horizontal artifacts) and score 3 (large consolidation) across all the models, except for VGG-16 and VGG-19. This suggests that for most of the networks, it is easier to differentiate the patterns associated with scores 0 and 3, but more challenging to differentiate scores 1 and 2, and this trend is similar as observed in a previous study [5]. However, VGG-16 and VGG-19 struggle to distinguish between score 2 (small consolidation) and score 3 (large consolidation) because both can appear similar in texture, making it hard for pretrained models to detect size differences. The overall results revealed that ResNet-18 achieves over 60% true positive rates for scores 1–3, highlighting the model’s effectiveness in capturing distinct features. Fig. 8 illustrates the performance of the DL models using Trainset-2. The confusion matrix indicates that all the models, except for DCNN, demonstrated superior

performance in classifying patterns associated with score 0 (horizontal artifacts) and score 2 (small consolidations). The ResNet-18 model was evaluated in two distinct configurations: first, it was trained using the transformation strategies outlined in [25], referred to as ResNet-18. In the second configuration, the transformation applied during training was removed, represented as ResNet-18^a. The results highlight that while the models performed adequately for scores 0 and 2, they struggled to accurately classify the patterns associated with scores 1 and 3. This suggests that incorporating affine transformations with the ResNet-18 model, which previously outperformed Trainset-1, reduces the model’s performance. One potential reason is that deeper networks may have difficulty learning the variations introduced by affine transformations, which can hinder their ability to distinguish the specific patterns associated with scores 1 (vertical artifacts) and 3 (large consolidation). This highlights that the model is confused between the patterns

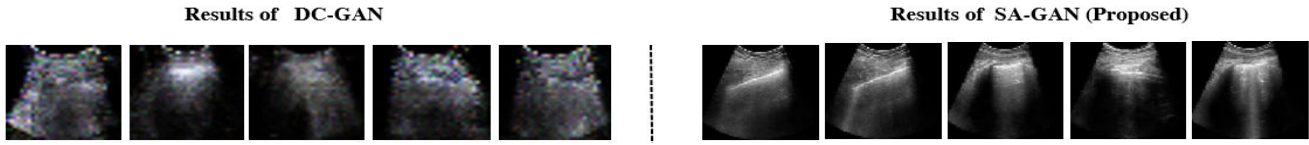


Fig. 9. Comparison between our SA-GAN approach and the state-of-the-art DC-GAN approach for data augmentation.

associated with scores 0 and 1, as well as scores 2 and 3, ultimately reducing classification performance.

C. Comparison of SA-GAN With Other GAN

A comprehensive comparison is conducted between our approach SA-GAN with the widely recognized state-of-the-art method, DC-GAN [26], [27]. This comparative analysis aimed to evaluate how effectively our approach tackles the specific challenges related to augmenting data in minority classes. One major challenge that we encounter is the quality of images generated by DC-GAN. The generated images exhibited significant issues, including a high level of blurriness and notable noise artifacts, despite its DC architecture. The model's struggle to capture and preserve fine details resulted in images that lacked the sharpness and clarity of LUS images. Furthermore, the presence of noise artifacts in the generated images is a significant concern that needs to be reduced. However, even after extensive training, there is a notable lack of substantial improvement in the quality of the generated images (see Fig. 9). In contrast, our proposed approach demonstrated an impressive ability to address this issue, producing more balanced results across different classes. DC-GAN, on the other hand, struggled to mitigate class imbalance, leading to an overrepresentation of majority-class images in its output.

D. Evaluation of a Cost-Sensitive Loss Function for Multiclass Classification

The best-performing models in all the three cases underwent further training with a cost-sensitive loss function for multiclass classification [28] to mitigate the issue of predictions deviating significantly from reference standards. For this reason, the models were retrained using these loss functions in the cases of Trainset-1 and the no-augmentation scenario. The best-performing model for Trainset-1 and the no-augmentation case was ResNet-18, while for Trainset-2, VGG-16 achieved the highest performance. After applying the cost-sensitive loss function, the overall accuracy of ResNet-18 improved slightly to 0.59, indicating a minor enhancement compared with using a standard loss function. However, in the case of Trainset-2, the performance of VGG-16 slightly declined after applying the cost-sensitive loss, resulting in an accuracy of 0.52. In the no-augmentation scenario, ResNet-18 achieved a similar performance, with an accuracy of 0.65. The detail results are given in the following link (see https://drive.google.com/file/d/11QbF9OfQc4cWVQ2uWlSP9f3tOF8dMAcp/view?usp=drive_link). This suggests that the effectiveness of cost-sensitive loss functions may depend on both the model architecture and the dataset characteristics.

IV. DISCUSSION AND CONCLUSION

In this study, we explored two strategies for class balancing. The first strategy used an advanced generative AI model, SA-GAN, while the second strategy used a classic transformation method known as affine transformation. Since GAN-based data augmentation has not been widely used in the state-of-the-art research for LUS imaging, our primary objective was to determine whether this model could effectively generate meaningful samples for the minority classes. To achieve this, we first developed the SA-GAN architecture intending to generate unique samples in underrepresented classes.

Initially, we used a smaller training set (Dataset-v1), consisting of 5500 samples for the training of the SA-GAN model. As a result, 900 samples were generated in the minority classes. In addition, we performed statistical analyses based on similarity measures and a blind review by clinicians, which indicated that the generated samples were unique and indistinguishable from real ones. After validating the effectiveness of SA-GAN on the smaller dataset, we proceeded to train the model using the larger multicenter dataset, ICLUSDB-V1, referred to as Dataset-v2, which contains 48 215 images in the training set. The SA-GAN was trained through multiple iterations to produce synthetic images for the minority classes, helping to balance the distribution of the training set. Consequently, 18 433 samples were generated for the minority classes, ensuring that each class contains 16 662 images, collectively referred to as Trainset-1. In addition, we implemented classic transformations to balance the training set by augmenting the minority classes, which we referred to as Trainset-2.

To develop an AI solution for the automated interpretation of LUS patterns, we later used DL methods, including DCNN, DCDM, and pretrained models such as VGG-16, VGG-19, and the state-of-the-art ResNet-18 [25]. These models were trained simultaneously on Trainset-1 and Trainset-2. Overall, the results presented in Table IV show that both the augmentation techniques yield competitive performance across different models. In Trainset-1, ResNet-18 stands out with an accuracy of 0.58, demonstrating its capability to effectively learn from the synthetic images generated by the SA-GAN. VGG-16 and VGG-19 also performed well with accuracies of 0.51 and 0.52, respectively. However, the other models, DCDM and DCNN, exhibited lower performance, highlighting the challenges in leveraging these architectures for this specific dataset.

In Trainset-2, which used classic transformation methods, VGG-16 achieved an accuracy of 0.58, demonstrating strong performance. This accuracy is comparable to ResNet-18's performance in Trainset-1. However, ResNet-18 exhibited a slight decrease in performance when applied to Trainset-2. The results demonstrate that both SA-GAN-generated images and

TABLE IV

PERFORMANCE COMPARISON OF DL CLASSIFIERS ON SA-GAN-AUGMENTED IMAGES (TRAINSET-1) AND TRADITIONAL TRANSFORMATION-AUGMENTED DATA (TRAINSET-2) WITHIN DATASET-V2. "NO AUGMENTATION" REFLECTS THE RESULTS OF TRAINING ON DATASET-V2 WITHOUT APPLYING ANY AUGMENTATION OR BALANCING THE DATASET DISTRIBUTION

Augmentation Technique	Models	Accuracy	Precision	Recall	F1 Score
Trainset-1	DCDM	0.50	0.49	0.50	0.50
Trainset-1	DCNN	0.43	0.41	0.43	0.43
Trainset-1	VGG-16	0.51	0.53	0.51	0.52
Trainset-1	VGG-19	0.52	0.53	0.52	0.52
Trainset-1	ResNet-18	0.58	0.59	0.58	0.58
Trainset-2	DCDM	0.53	0.50	0.53	0.53
Trainset-2	DCNN	0.47	0.47	0.47	0.47
Trainset-2	VGG-16	0.58	0.59	0.58	0.58
Trainset-2	VGG-19	0.52	0.49	0.52	0.52
Trainset-2	ResNet-18	0.57	0.57	0.57	0.57
Trainset-2	ResNet-18 ^a	0.55	0.54	0.55	0.55
No Data Augmentation	DCDM	0.50	0.52	0.50	0.50
No Data Augmentation	DCNN	0.43	0.42	0.43	0.43
No Data Augmentation	VGG-16	0.48	0.48	0.48	0.48
No Data Augmentation	VGG-19	0.52	0.50	0.52	0.52
No Data Augmentation	ResNet-18	0.66	0.72	0.66	0.65

classic transformation-augmented data can lead to competitive performance. Therefore, understanding the characteristics of each model and how they interact with different augmentation techniques is essential for optimizing classification performance in LUS imaging and other medical domains.

The results in Table IV demonstrate that training on the entire dataset without augmentation achieves robust performance. Our purpose in applying two augmentation techniques SA-GAN and traditional augmentation was to explore whether synthetic data generation could enhance performance in this context, an approach not yet used in the previous state-of-the-art research on LUS image classification. While our findings indicate that augmentation does not improve performance on this dataset, as confirmed by the cost-sensitive loss function, they establish a valuable benchmark for future research. They provide an important benchmark for future studies. This study establishes the groundwork for understanding when and how synthetic data generation could be beneficial in medical imaging.

V. LIMITATIONS AND FUTURE WORK

GANs hold promise for data augmentation but come with certain limitations. GAN-based data generation is computationally expensive and time-intensive, as noted in other studies [29]. In this work, the GAN training process spanned three months and produced images at a resolution of 64×64 , which aligns with resolutions used in many state-of-the-art studies [30], [31]. However, the generated images exhibited some blurriness, indicating room for improvement. Higher resolution image generation would necessitate significantly greater computational resources [32], which could limit the accessibility of such methods.

Furthermore, the generalizability of classifiers developed using the proposed techniques remains a significant challenge. This is highlighted in a study by Khan et al. [33], which demonstrated that the cross-hospital performance of a ResNet-18 model trained on LUS data was considerably poor. In their study, the model was trained on a dataset from Brescia ($N = 13$) and evaluated on data from Rome. Despite using various training strategies, such as adjusting the size of the training

dataset, the model struggled to generalize effectively to the target hospital data. As part of future work, we plan to explore domain adaptation through transfer learning and fine-tuning, where pretrained models on a source domain are adapted to a target domain with minimal labeled data. In addition, adversarial domain adaptation techniques, such as domain adversarial training, could help in reducing distributional shifts between different datasets. We also aim to investigate feature alignment methods such as maximum mean discrepancy (MMD) [34] and CORrelation ALignment (CORAL) [35] to enhance feature transferability. Furthermore, self-supervised learning and contrastive learning approaches may be leveraged to improve model robustness across diverse domains. These adaptations can potentially mitigate domain shifts and enhance classification performance in real-world applications. Another limitation lies in the reliance on clinician interpretations of LUS patterns. The subjective nature of visual scoring introduces variability and inconsistency in label assignments, which could lead to biased classification results. This highlights the need for standardized and objective approaches to label LUS data, ensuring more reliable model training and evaluation. Looking forward, advancements in generative AI offer promising opportunities to address these limitations. Improving image resolution and quality remains a key focus, with plans to design networks capable of handling larger input sizes. This would enhance the suitability of generated images for clinical applications. In addition, refining the SA-GAN architecture, incorporating advanced loss functions, and exploring diffusion models may help reduce blurriness, expedite the generation process, and improve overall performance.

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